

CAPE CHARLES 5 ENE, VA, USA

WMO: 745900

Lat: 37.2907N

Lon: 75.9270W

Elev: 29

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	23.7	2.4	6.2	20.5	7.4	8.0	24.8	22.4	29.8	19.7	29.1	8.8	N/A	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	90.2	76.9	87.7	76.2	85.8	75.5	79.6	86.0	78.4	84.4	77.4	83.1	5.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	145.4	82.7	76.8	139.9	81.7	75.7	134.6	80.8	43.1	86.0	41.8	84.4	40.8	83.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.0	14.7	12.7	DB	15.1	94.8	4.6	2.8	11.9	96.9	9.2	98.5	6.6	100.1	3.3	102.2	
			WB	11.6	81.1	4.9	1.7	8.1	82.4	5.2	83.4	2.4	84.3	-1.2	85.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	39.3	41.0	46.8	57.0	66.1	74.1	78.3	76.6	71.8	61.9	50.8	44.3	(6)
(7)		DBStd	15.69	9.74	9.31	8.99	8.33	7.20	5.57	4.32	4.22	5.45	7.65	7.89	8.75	(7)
(8)		HDD50	1145	353	280	174	25	0	0	0	0	4	84	225		(8)
(9)		HDD65	3614	797	672	565	266	78	4	0	10	152	427	643		(9)
(10)		CDD50	4469	22	28	77	234	499	724	877	825	653	374	109	47	(10)
(11)		CDD65	1462	0	0	2	25	112	278	412	360	213	56	3	1	(11)
(12)		CDH74	10821	0	0	10	123	529	2028	3773	2955	1205	195	2	1	(12)
(13)		CDH80	2807	0	0	0	15	98	552	1167	750	205	20	0	0	(13)
(14)	Wind	WSAvg	5.6	6.6	6.6	7.1	6.6	5.5	4.8	4.3	4.2	5.0	5.2	5.7	5.7	(14)
(15)	Precipitation	PrecAvg	45.5	3.1	3.0	4.1	3.4	3.3	3.9	4.9	4.8	4.6	3.9	3.0	3.7	(15)
(16)		PrecMax	61.1	6.9	7.4	9.6	5.0	7.1	8.7	10.2	11.0	11.6	10.1	7.8	8.1	(16)
(17)		PrecMin	34.4	1.7	0.5	0.4	0.9	0.7	1.7	0.6	1.7	0.7	0.0	0.2	1.1	(17)
(18)		PrecStd	7.1	1.1	1.5	1.7	1.1	1.6	1.6	2.3	2.3	2.9	2.2	1.9	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	66.9	70.6	75.1	82.3	86.4	92.4	94.6	92.2	87.6	82.3	72.9	70.8		(19)
(20)		MCWB	61.9	62.9	63.9	69.5	72.8	76.0	77.8	77.9	76.0	72.4	65.8	66.2		(20)
(21)		DB	61.8	65.0	69.5	77.7	82.8	88.8	90.8	88.5	85.0	78.6	68.7	66.0		(21)
(22)		MCWB	55.2	58.5	59.1	65.3	70.9	74.7	77.7	77.0	75.4	71.8	64.0	62.0		(22)
(23)		DB	58.0	60.4	65.2	73.6	79.7	85.8	88.2	86.3	82.3	76.0	65.7	61.9		(23)
(24)		MCWB	52.4	54.5	58.8	65.3	70.4	73.9	77.2	76.2	74.6	71.0	61.3	58.4		(24)
(25)		DB	54.1	55.3	61.1	69.4	76.8	83.1	86.1	84.2	80.0	73.3	62.8	57.7		(25)
(26)		MCWB	50.0	50.5	55.6	61.8	69.3	73.2	76.4	75.2	73.6	68.3	58.4	55.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	61.9	64.3	64.9	71.9	75.8	78.3	82.0	80.5	78.5	76.2	67.9	67.9		(27)
(28)		MCDWB	65.9	69.0	71.4	77.0	81.5	84.5	87.6	85.7	83.0	79.2	70.4	70.8		(28)
(29)		WB	56.4	60.9	62.1	68.8	73.4	77.0	80.3	79.2	77.4	74.6	65.3	63.6		(29)
(30)		MCDWB	59.0	65.4	67.1	74.2	79.8	84.3	86.7	84.9	81.5	77.7	68.3	66.7		(30)
(31)		WB	53.0	56.9	60.0	66.1	72.1	75.7	79.0	78.0	76.5	72.6	62.6	60.2		(31)
(32)		MCDWB	55.7	60.3	64.8	71.6	77.7	82.5	85.9	83.2	80.3	75.6	65.3	62.7		(32)
(33)		WB	49.8	52.0	56.7	63.5	70.8	74.8	77.9	76.9	75.3	69.2	59.6	56.4		(33)
(34)		MCDWB	52.6	55.7	60.8	68.6	75.9	81.1	84.5	81.7	79.0	72.5	62.8	58.6		(34)
(35)	Mean Daily Temperature Range	MDBR	13.4	14.3	14.6	15.3	13.7	13.4	13.1	12.6	12.5	13.6	14.1	13.9		(35)
(36)		5% DB	18.1	20.4	20.1	19.7	16.4	15.9	15.0	14.2	14.2	14.0	15.6	17.1		(36)
(37)		MCWBR	17.3	15.3	13.0	10.0	8.5	6.9	6.5	6.1	6.8	7.8	11.5	13.9		(37)
(38)		5% WB	17.7	20.0	17.8	16.3	13.9	13.9	14.0	12.3	11.1	11.3	14.1	16.7		(38)
(39)		MCWBR	17.2	16.6	13.2	10.3	8.3	7.0	6.8	6.6	6.2	7.4	12.8	14.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.320	0.346	0.393	0.436	0.482	0.513	0.507	0.412	0.355	0.325	0.314		(40)
(41)		taud	2.547	2.515	2.474	2.353	2.266	2.186	2.114	2.118	2.376	2.529	2.561	2.568		(41)
(42)		Ebn at Noon	280	291	292	282	270	257	247	246	266	274	272	271		(42)
(43)		Edh at Noon	24	28	32	39	43	47	49	48	36	28	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	722	971	1312	1669	1886	2040	1952	1764	1436	1121	819	641		(44)
(45)		RadStd	59	100	103	115	169	136	93	100	118	117	67	51		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (61 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	+284	+185	(47)

Nomenclature: See separate page